

Homework 11 - FIR filters

Deadline: 4th test date (4th of June)

The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 50 points.

1. What do we mean by a finite impulse response (FIR) filter? How does it differ from IIR filters in terms of their linear constant-coefficient difference equation?
2. Compare FIR and IIR filters.
3. Explain the advantages and disadvantages of FIR filters compared to IIR filters. In what scenarios would you prefer one over the other?
4. What is the Fourier/window method for designing FIR filters? How does it work and what are its key benefits?
5. Implement a Python script to compute and plot the frequency response of the FIR filter designed in the previous question.
6. Explain the Fourier/window method for designing high-pass, band-pass, and band-stop FIR filters. Provide examples of the window functions used in each case.
7.  Prove that a symmetric FIR filter of odd length has a constant group delay. What is the value of this delay?
8. Why does truncating the ideal sinc impulse response introduce ripples in the frequency response?
9. Explain why an ideal low-pass filter cannot be both causal and finite length.
10. Derive the impulse response of an ideal high-pass FIR filter.
11. A FIR filter has coefficients

$$h[n] = [1, 2, 3, 2, 1].$$

Is the filter linear phase?

12. Why are FIR filters always BIBO stable?
13. Why does increasing the filter order improve the sharpness of the transition band?
14. What happens if the FIR coefficients are not symmetric?
15. Prove analytically that the zeros of the polynomial $P(x) = \sum_{n=0}^N x^n$ are uniformly distributed around the unit circle in the complex plane apart from $x = 1$. Show it also numerically using the `numpy.roots` method.
16. Compute and plot the response of a 21-point moving-average filter ($\frac{1}{N} \sum_{k=0}^{N-1} x[n-k]$).
17. Argue that the moving average filter acts as a low-pass filter.

18. Represent the frequency response of the moving average filter and establish the qualitative dependence of the cutoff frequency and roll-off on the length of the window.
19.  Study analytically the gain at the Nyquist frequency for the moving average filter.
20.  Study analytically the dependence of the cutoff frequency on the length of the moving window for very large window size.
21. Why is the ideal response usually specified in the frequency domain first?
22. Consider

$$h[n] = \delta[n] - \delta[n-1].$$

What type of filter is this: FIR/IIR, low-/high-/passband? Justify!

23. Why does a linear-phase FIR filter delay all frequency components equally?
24. Explain the difference between filter order, number of coefficients and impulse response length.
25. Design a lowpass FIR filter of order 20 with a cutoff frequency of 84kHz and a sampling frequency of 20 kHz using the Fourier/window method. Represent the frequency response and comment on the gain and phase relationship.
26.  Same task as above but for a highpass filter with a cutoff frequency of 8kHz.
27.  Same task as above but for a bandpass filter with passband of 4kHz- 8kHz.
28. Show that, to normalize the DC gain of a low-pass FIR filter, the coefficients $b_{\text{low}}[n]$ should be divided by
29.  Show that, to normalize the Nyquist-frequency gain of a high-pass FIR filter, the coefficients $b_{\text{high}}[n]$ should be divided by

$$\left| \sum_{n=0}^N (-1)^n b_{\text{high}}[n] \right|.$$

30.  Show that, to normalize the gain of a band-pass FIR filter at the center frequency f_0 , the coefficients $b_{\text{band}}[n]$ should be divided by

$$\left| \sum_{n=0}^N b_{\text{band}}[n] e^{-j2\pi f_0 n / f_s} \right|, \quad f_0 = \frac{f_l + f_h}{2}.$$